E S S [729] E S S

Effaying weighed, and made of paper not impregnated with alum. It is to be dried at frift with a gentle heat, but afterwards expelled for five minutes to a beat of 10:00 On cooling, it to the weights together with the fiter; the brown weight of which mult afterwards be fobricked. The fedmena is both included in bottle for a filter than once impregnated with false matter cannot be freed from it again without great difficulty, especially if an internal of one burns intervents. The alkali mute ufc of in Mr Bergantis experiments, was that of fols faturated with aerial acid, also ploughedicated skills needed by defibrating equal weights of poor mice and create on tartrationally mixed together; the reifulaum is the common white flux. Half an onces of this is indifferent in half a quadrans of distilled water. This is the adda, as a digital heat, two ounces of Prulfian blue, carefully avoiding (ch) an evervidence upon throw any thing or, which cellily happen if the quantity be too large. The pig-ment footo lodes its beautiful blue colour, growing not red but blacks, which shows that a decomposition has taken place. The Prussian blue rule of in his expert mentioned in 100 parts only 33 of the pigment and 57 of the earth is low. If we employ the blue male exact way alum, 22 grams of K will saturate the half onces of alkaline fat more completely than the two causes of the final already described. But in whatever manner the operation is performed, after the addition of the hall quantity, the whole mult to ex-paid to a stronger diguling heat, and differed with wooden faults. If the liquor be too much diminishely on proportion, the deficit mult be figured by adding more water. When the liquor becomes clear the réiduum mult be collected upon filtering paper, and gradually washed with warm water until all the fullable part is intended, where, if the proportion has been properly conducted, the filtered liquor amounts to a whole quamings, of a brown yellow color, and fo well futurated with colouring matters, that it does not change the colour of paper tinged with Brafl wood. This historic, however, contains a funqu-tivity of Prulfian blue, about 3d, No a ctwy of the al-kali. Thefe should be previously farparted by an acid, or, which is better, corrected by flutstanding from the weight of the delinattor of clay grounds for each quasrans of the likivium. When we with to examine the colour of the products exactly, however, the likivium we employ mult mustically be well departedly, for by neglecting this precaution we may easily perturbeour-fleew that any mist precipitated by the likivion has a blue color. When we out with to ascertain the weight, the likivium, having the final proportion of Prulfian blue becomes, may be greatest, but fill the proper correction mult ultimately do made ufc of; for the precipitating said is wont to impair the quali-tion of the likivium, and even to deliver them altogether, especially in a warm temperature. Calcorous cards, whether in its mild or canific fixate, is also capable of ablifting a coloured fuhlance from iron and other metals. Int the precipitation of metal the metal, it is to be ordered, that the card of the foliation ought to be somewhat predominant; but any confidable excess mult be corrected occasionally, either by alkali, water, or spirit of wine. In the following experiments an effay ewt. was always employed, unlels where it is expressly mentioned otherwise; conclusions sufficiently accurate may in- dred be obtained from 37th, my functions from used in Mr smaller quantities. In their cafes our author mentions Respond a the initial quantity, applying to them thoir formula of reperi-calculation which are founded on the mutual purpose results. tons of the proximate principles confluiting metallic fails. By us early fuhlition, one time broke may be useful by those who employ 5 or 4 vet. We now come to defer the method of elifying the ones of the partities mental. 1. _Ores of Gold._ This metal occurs in the bowels of the earth native, poffelling a complete metallic forms of the area although in general the small particles of it are for in-of gold. Forperfected various mattons that they are entirely invisible. It is also found mineralized, or united with fulpharby means of iron or from other me- inclined. Therefore ten types of cost we feel counterly fe-partably. Native gold is very felddom, if every, free from hote- Native rogreous matters; the molt uful mixtures are cop-gold, per, silver, and formtimes iron. The first of the remaining the medium, and may be particularly calefed by diffusing the gold in auguegies, and precipitating by by martial virtvel; the second falls during the following yielding a flatelry, which, being washed and dried, allows the weight of the fiber cou-tained, and the iron may be discovered by pholgills- coated alkali. The precipitate occurred by martial vis-fold in pure gold in its metallic base, but very definitely divided, and therefore its weight requires so correction. Hence it appears how small a portion of gold in herrent in the ones of other metals may be extracted; besides, it follows containing the molt minute parcle of gold initially produces the purple precipitate of Caffius, with a foliation of its property pre-parallel. As to the ore which contains

gold adhering to and Ores of this furrounded by flony particles, r. We mult reduce a deter-metal pro-mined weight to an impulsive powdery texture and pretty da-alitarian. Then let the powdery weighted forvolume, for boiled in aqua regis, as long as any thing is taken, by the mediums), after which, let the exhausted ore, well washed, be collected, refecuted to spinism, and weighed. Let the clear features (the colour of which, in forens above affords a method of judging) be precipitated in the natural way by martial virtvel; the precipitate well washed, dried, and weighed, flows the gold, which, added to the weight of the exhausted ore, might to be equal to the original weight, unless fenome what has been dispersed by the pulverification, or beliefs force of the mattons has entered the meltdown. The former of thefts delivered by comparing the weights before and after pulverification; the latter by precipitants. When grains of gold are mixed with loofe earthy particles, they are sometimes easily ferated by mec- channel application of water. When the metal is mineralised by fulphur, as in the golden pyster, let too an more effay ctw, reduced to arilized by powder be gently boiled in the nitrated acid, or rather fulphur, digested in a heat of 5080ř, left the fulphur should be delayed. It is even neceffary to employ a more gentle brat for this purports, that the fulphurcous pr-ticles

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